

Cancellation-aware order coherence: an exact decomposition of absolute concordance change into signed Systemic Tau and a cancellation index

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Abstract

Systemic Tau (τ_s), the mean pairwise Kendall concordance of module activity on a common event index, is a signed average: concordance gained on some pairs can offset concordance lost on others, so structural change can pass unseen. A cancellation-immune alarm is the mean absolute change D of pairwise concordance from its calibration baseline over a fixed edge set. Here we show that both statistics are instances of one averaging operator on the pairwise concordance matrix and that, on a common edge set, D decomposes exactly into the absolute change of the signed mean plus a non-negative cancellation term equal to twice the minority-sign mass of edge changes. The resulting index $\kappa \in [0,1]$ is the fraction of total change that the signed mean cannot see. We derive bounds, invariances and the no-change behaviour of D and κ , neither of which is near zero without change, and give a conformal alarm rule with a finite-sample false-alarm guarantee. In synthetic event streams, balanced rewiring left τ_s at its false-alarm level while D detected it in 95.0% of windows.

Systemic Tau (τ_s) summarizes the coordination of a multivariate system as the mean of pairwise concordances between the activity streams of its modules¹⁻³, each computed as Kendall's tie-corrected coefficient τ_b on a common event index^{4,5}. In the companion article⁶ it serves as the essential variable of a regenerative controller, in the sense of Ashby's ultrastable system⁷ and by analogy with the set points of homeostatic plasticity⁸. A signed mean has a structural blind spot, however. Networks that compute contain concordant and anti-concordant relations side by side, and a change that raises concordance on some pairs while lowering it on others can leave τ_s where it was. The companion article therefore makes the operative alarm a matrix statistic, the mean absolute change D of pairwise concordance from its calibration baseline over the edges present at the end of calibration, and retains τ_s as a signed control.

Two statistics raise three questions. How are they related? How much of the change registered by D is invisible to the signed mean? And how should D be thresholded, given that it is positive even when nothing has changed? Here we show that τ_s and D are two instances of one averaging operator on the pairwise concordance matrix; that on a common edge set D decomposes exactly into the absolute change of the signed mean and a non-negative cancellation term; that the ratio of the two defines a cancellation index with a direct interpretation; and that the no-change behaviour of both quantities,

which is far from zero, can be predicted and calibrated. A conformal rule gives the alarm a finite-sample false-alarm guarantee, and synthetic event streams illustrate each result.

One operator, two statistics

Let $\tau_b^{(p)}(t)$ be the concordance of pair $p=(i,j)$ in the window ending at t (Methods, equation (9)), let P be the set of all $N(N-1)/2$ pairs of N modules, and let the matrix $M(t)$ collect these values. Calibration on a reference period yields a baseline B_p for each pair, the mean of $\tau_b^{(p)}$ over n_{cal} calibration windows, and a set of edges $E \subseteq P$ fixed at the end of calibration. In the companion controller E contains the pairs joined by a connection of the network, but nothing below depends on how E is chosen, only on its being fixed before monitoring begins. For any set of pairs S and any function f of a current value x and a baseline B , define

$$F_t(S, f) = \frac{1}{|S|} \sum_{p \in S} f(\tau_b^{(p)}(t), B_p), \quad B_p = \frac{1}{n_{\text{cal}}} \sum_{w=1}^{n_{\text{cal}}} \tau_b^{(p)}(w) \quad (1)$$

Systemic Tau and the operative alarm are two instances of F , as is the signed change on the edge set:

$$\tau_s(t) = F_t(P, x), \quad D_E(t) = F_t(E, |x - B|), \quad \bar{\Delta}_E(t) = F_t(E, x - B) = \tau_{s,E}(t) - \tau_{s,E}^{\text{cal}} \quad (2)$$

where $\tau_{s,E}$ is the signed mean restricted to E and $\tau_{s,E}^{\text{cal}}$ its calibration mean; the last equality holds because the baselines are calibration means and F is linear in f . The instances differ in two choices only: the set of pairs, which determines which relations are monitored, and the function, which determines whether the sign of a change is retained. What they share is the pairwise matrix $M(t)$. Statistics of ordinal structure beyond pairs, which the theory of hierarchical ordinal conjunctions addresses^{2,3}, are not in general functions of $M(t)$ and lie outside this operator.

An exact decomposition

Write $\Delta_e(t) = \tau_b^{(e)}(t) - B_e$ for the change on edge e , and let S_+ and S_- be the total magnitudes of the positive and of the negative changes on E . Because the signed mean averages the same changes whose magnitudes D_E averages,

$$D_E = |\bar{\Delta}_E| + C_E, \quad C_E = \frac{2}{|E|} \sum_{e \in E^-} |\Delta_e| = \frac{2}{|E|} \min(S_+, S_-) \geq 0 \quad (3)$$

where E^- is the set of edges whose change has the sign opposite to that of $\bar{\Delta}_E$ (Methods, Proposition 1). The first term is the part of the change that the signed control sees; the second, C_E , is the part it loses to cancellation, twice the mass of the minority sign. The triangle inequality guarantees $C_E \geq 0$, and the cancellation index

$$\kappa_E = \frac{C_E}{D_E} = \frac{2 \min(S_+, S_-)}{S_+ + S_-} = 1 - \frac{|\bar{\Delta}_E|}{D_E} \in [0, 1] \quad (4)$$

is the fraction of the total absolute change that is invisible to the signed mean. It is zero when all edges change in the same direction, in which case D_E and $|\bar{\Delta}_E|$ coincide, and one when positive and negative changes balance exactly, in which case the signed control is blind however large D_E may be.

The pair $(|\Delta_E|, C_E)$, or equivalently (D_E, κ_E) , is therefore a two-coordinate description of the change with no free parameter: lines of constant D_E are anti-diagonals of the $(|\Delta_E|, C_E)$ plane, and the position along such a line is fixed by κ_E (Fig. 1a).

The decomposition also explains why scalar combinations of the two statistics are not used. A statistic such as $D - \lambda|\Delta\tau_s|$, or a rule that fires on D only while τ_s stays within its band, subtracts from D the coherent component that D already contains. Such constructions suppress the alarm precisely when the change is global and obvious, and their weight has no calibration principle. Equation (3) makes them unnecessary: the information on which they draw is carried, without a weight, by the coordinates $(|\Delta_E|, C_E)$.

The identity is exact when the signed control is computed on the same edge set as D_E . The Systemic Tau of the companion article averages over all pairs, of which E is a subset of relative size $\rho = |E|/|P|$. Its change from calibration, $\Delta_P = \tau_s(t) - \tau_s^{\text{cal}}$, mixes the signed changes on and off the edge set, and the decomposition acquires a remainder:

$$\bar{\Delta}_P = \rho \bar{\Delta}_E + (1 - \rho) \bar{\Delta}_{P \setminus E}, \quad D_E = |\bar{\Delta}_P| + C_E + R, \quad |R| \leq (1 - \rho) |\bar{\Delta}_{P \setminus E} - \bar{\Delta}_E| \quad (5)$$

The remainder vanishes when $E=P$ and is bounded by the disagreement between the signed changes on and off E . The all-pairs τ_s can therefore move while every monitored edge is unchanged, and it can stay still while the edge set cancels; we recommend reporting the edge-restricted control $\tau_{s,E}$, for which equation (3) holds exactly, alongside τ_s over all pairs.

Because every τ_b lies in $[-1, 1]$, $0 \leq |\Delta_E| \leq D_E \leq 1 + |E|^{-1} \sum_e |B_e| \leq 2$ and $0 \leq C_E \leq D_E$. Being rank-based, like ordinal-pattern measures⁹, all these quantities are invariant to strictly increasing transformations of any stream and to relabelling of modules. Only D_E is also invariant to reversing the polarity of a module, for instance to the choice of which of two complementary codes counts as active: reversal negates every concordance involving that module, and with it the signs of its edge changes, which alters $|\Delta_E|$, C_E and κ_E but not their sum (Methods). The split between coherent and cancelled change is thus defined relative to the sign convention of each stream; D_E is not.

Behaviour without change and a calibrated alarm

Neither D_E nor κ_E is near zero when nothing has changed. Each Δ_e then carries the estimation error of the current window and of the baseline, with a standard deviation σ_e that follows from Kendall's null variance with ties⁵, the factor $1 + 1/n_{\text{cal}}$ for the baseline, and a correction $f_e \geq 1$ for serial correlation of the binned streams¹⁰ (Methods, equations (12) and (14)). For approximately Gaussian errors that are uncorrelated across edges, as for mutually independent modules,

$$E_0[D_E] \approx \sqrt{\frac{2}{\pi}} \frac{1}{|E|} \sum_{e \in E} \sigma_e, \quad E_0[|\bar{\Delta}_E|] \approx \sqrt{\frac{2}{\pi}} \frac{1}{|E|} \left(\sum_{e \in E} \sigma_e^2 \right)^{1/2}, \quad \kappa_0 \approx 1 - \frac{\left(\sum_e \sigma_e^2 \right)^{1/2}}{\sum_e \sigma_e} \quad (6)$$

and $\kappa_0 \approx 1 - |E|^{-1/2}$ when the σ_e are similar. The signed mean averages estimation noise away at the rate $|E|^{-1/2}$, whereas the mean absolute value does not, so that without change κ_E approaches one as the edge set grows. A large κ_E is evidence of cancellation only when D_E itself exceeds its calibrated threshold. The same asymmetry governs sensitivity. For a coherent shift δ on every edge with Gaussian error of standard deviation σ , the expected absolute change is that of a folded normal variable¹¹,

$$E|\Delta_e| = \sigma h\left(\frac{\delta}{\sigma}\right), \quad h(u) = \sqrt{\frac{2}{\pi}} \exp\left(-\frac{u^2}{2}\right) + u(2\Phi(u) - 1) = \sqrt{\frac{2}{\pi}} \left(1 + \frac{u^2}{2}\right) + O(u^4) \quad (7)$$

so that near the noise floor D_E grows quadratically in δ while the signed mean grows linearly, and for δ small relative to σ the signed control detects coherent change earlier (Methods). D_E , by contrast, detects balanced change, which the signed control cannot detect at all, and responds most strongly to change concentrated on a few edges.

These properties determine the alarm rule. The threshold for D_E must come from the no-change distribution of D_E itself, not from zero and not from a band around τ_s . We use the calibration windows as a conformal reference¹²: each calibration window w receives a score s_w , its mean absolute deviation from the mean of the other windows (Methods, equation (15)), and

$$p_D(t) = \frac{1 + \sum_{w=1}^{n_{\text{cal}}} 1[s_w \geq D_E(t)]}{n_{\text{cal}} + 1}, \quad \text{alarm} \Leftrightarrow p_D(t) \leq \alpha, \quad e^* = \arg \max_{e \in E} |\Delta_e(t)| \quad (8)$$

fires with probability at most α when the windows are exchangeable, as they are for an unchanged stationary process with nearly independent windows (Methods). Each alarm is reported with its diagnostic pair $(|\Delta_E|, C_E)$, or (D_E, κ_E) , and with its source e^* , the edge that contributes most to D_E . No-dependence references, circular-shift surrogates^{13–15} and rate-matched independent streams, keep the role they have for τ_s ⁶: they establish that the baselines differ from what independent streams would produce, and they supply the noise floor of equation (6) against which D_E and κ_E are read. Where weak coherent change must also be caught, the signed control on E can be monitored with its own conformal band and the alarm raised when either p-value falls to $\alpha/2$ or below, a union of two calibrated tests that keeps the false-alarm probability at most α without introducing a weight.

Numerical illustration

We simulated binned event streams from $N=10$ modules whose firing probabilities are modulated by latent binary states, one per edge of a fixed coupling graph with $|E|=16$ edges among the 45 pairs, half concordant and half anti-concordant (Methods, equation (16)). Each replicate was calibrated on 39 windows and then monitored for five windows under one of four conditions: no change; a coherent shift, produced by a common drive that raises every pairwise concordance; balanced rewiring, in which four concordant and four anti-concordant edges invert their polarity; and independent streams matched in rate and autocorrelation, used for calibration and monitoring alike. The calibrated baselines lay far from the no-dependence floor (mean $|B_e|$ of 0.124 against 0.016 for circular-shift surrogates).

Table 1 and Fig. 1 summarize 200 replicates. Without change, the three conformal alarms fired in 4.4–5.1% of windows in the coupled system and in 4.6–5.7% for independent streams, close to the nominal 5%. D_E was far from zero under both nulls: for independent streams its mean was 0.098, against 0.098 predicted by equation (6) and 0.098 from circular-shift surrogates; without the serial-correlation factor the prediction was 0.086. The mean of κ_E was 0.75 against a predicted 0.75. Balanced rewiring left both signed statistics at their false-alarm level (alarms in 3.6% of windows on E and 4.9% over all pairs), whereas D_E fired in 95.0%, with $\kappa_E=0.86$ (Fig. 1b); the edge with the largest change was a rewired edge in 98.3% of these alarms, against 50% expected by chance. Coherent change of similar mean magnitude lay mostly in the signed component ($\kappa_E=0.23$). It was detected by the signed control on E in 84.2% of windows and by the all-pairs τ_s , whose non-edge pairs shared the drive, in 99.1%, but by D_E in only 48.6%. Across magnitudes (Fig. 1c), D_E dominated for balanced change and the signed control for weak coherent change, as equations (6) and (7) predict; the union rule detected 78.9% of coherent and 88.7% of balanced windows.

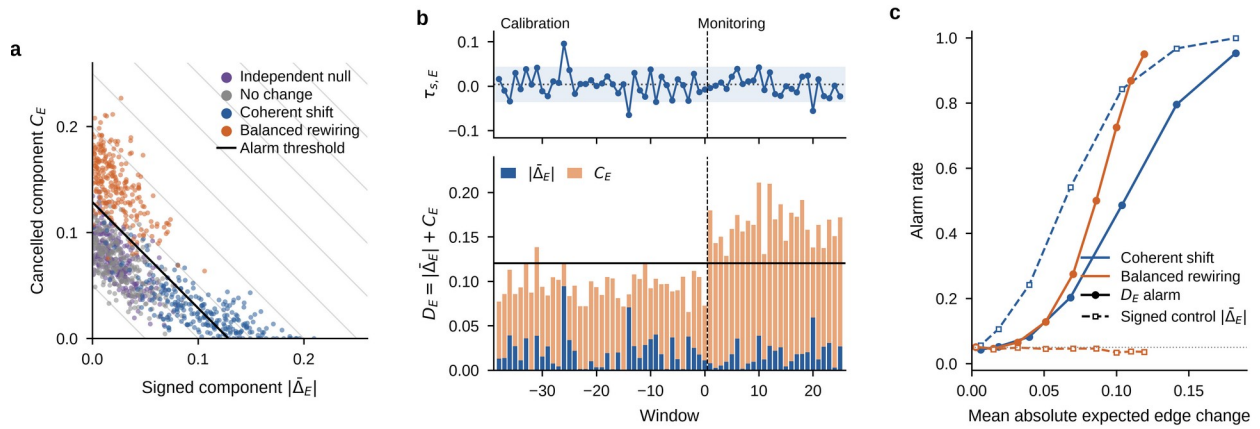


Fig. 1 | Decomposition of absolute concordance change in synthetic event streams. **a**, Monitored windows (250 randomly chosen per condition) placed by their signed component $|\Delta_E|$ and cancelled component C_E . Grey anti-diagonals are lines of constant $D_E = |\Delta_E| + C_E$; the black line is the median conformal alarm threshold without change ($\alpha=0.05$). Coherent change moves windows along the horizontal axis and balanced rewiring along the vertical axis; both null conditions lie near the origin with $C_E > |\Delta_E|$. **b**, One realization of balanced rewiring. Top, the signed control $\tau_{s,E}$ with its calibration mean (dotted) and alarm band (shaded). Bottom, D_E stacked as $|\Delta_E|$ (blue) plus C_E (orange), with its alarm threshold; calibration windows are shown against leave-one-out baselines. After the change (dashed line) the increase is carried almost entirely by C_E . **c**, Alarm rates of D_E (solid) and of the signed control (dashed) against the mean absolute expected edge change, for coherent shifts of increasing strength and for progressive polarity rewiring; each point summarizes 1,000 windows (200 replicates $\times$ 5 windows); dotted line, α .

Scope

The decomposition is an algebraic identity. It holds in every window and for every edge set, and it applies to any signed average of pairwise changes, whatever the association measure. Its statistical companions, the no-change predictions and the conformal rule, rest on approximations (Gaussian edge errors, weak correlation between edge errors, exchangeable windows) that the simulation supports but that must be checked in each application, which is why every threshold here is calibrated rather than fixed. The illustration is synthetic and small; it shows how the statistics behave when the ground truth is known and says nothing about any particular neural or artificial system.

Nor does it establish when cancellation occurs in practice. Early-warning signals for critical transitions¹⁶ are also read from system-level statistics, but whether balanced reorganization, and hence cancellation, accompanies such transitions is an empirical question that this note does not address. What the decomposition provides is an accounting: whenever D_E raises an alarm, it states how much of the change the signed Systemic Tau would have seen and how much it would have missed.

Table 1 | Decomposition and alarm rates in four synthetic conditions. Means over 1,000 monitored windows (200 replicates $\times$ 5 windows); $N=10$ modules, $|E|=16$ edges, 45 pairs. $\Delta\tau_s$, change of τ_s over all pairs; Δ_E , D_E , C_E , K_E , signed change, mean absolute change, cancelled component and cancellation index on E . Alarm rates use conformal thresholds at $\alpha=0.05$ from 39 calibration windows; the union rule fires when D_E or the signed control on E reaches $p \leq \alpha/2$. Source, fraction of D_E alarms whose largest edge change lies on a rewired edge (chance 50%). Wilson 95% intervals of all rates are within ± 3.1 percentage points.

Condition	Mean over windows					Alarm rate (%)				
	$\Delta\tau_s$	Δ_E	D_E	C_E	K_E	D_E	$\tau_{s,E}$	τ_s , all pairs	Union	Source (%)
No change	-0.000	0.001	0.095	0.070	0.74	5.1	5.1	4.4	3.6	—
Coherent shift ($\gamma=0.5$)	0.106	0.104	0.131	0.027	0.23	48.6	84.2	99.1	78.9	—
Balanced rewiring	-0.001	-0.001	0.167	0.144	0.86	95.0	3.6	4.9	88.7	98.3
Independent null	-0.000	-0.002	0.098	0.074	0.75	5.1	5.7	4.6	5.1	—

Methods

Pairwise concordance and baselines

Module i contributes a binned activity stream $z_i(k)$ on a common discrete event index, for example event counts in bins of b ticks. For bins $k < l$ of a window of W bins, Kendall's tie-corrected coefficient⁵ is the cosine between the sign vectors of two streams,

$$\tau_b^{(ij)} = \frac{\sum_{k < l} \xi_i(k, l) \xi_j(k, l)}{\left(\sum_{k < l} \xi_i(k, l)^2 \right)^{1/2} \left(\sum_{k < l} \xi_j(k, l)^2 \right)^{1/2}}, \quad \xi_i(k, l) = \text{sgn}(z_i(l) - z_i(k)) \quad (9)$$

which lies in $[-1, 1]$. It is set to zero when either stream is constant in the window, and the fraction of such undefined pairs is reported; it was zero throughout the simulations. The baseline of pair p is the mean of $\tau_b^{(p)}$ over the calibration windows (equation (1)); the mean, rather than a median, is what makes Δ_E equal to the change of $\tau_{s,E}$ from its calibration mean. The edge set is fixed when calibration ends and is not updated during monitoring, so that edges lost after calibration remain monitored and edges gained are not.

Proposition 1

For real changes Δ_e , $e \in E$, with S_+ and S_- defined by

$$S_+ = \sum_{e: \Delta_e > 0} \Delta_e, \quad S_- = \sum_{e: \Delta_e < 0} |\Delta_e|, \quad D_E = \frac{S_+ + S_-}{|E|}, \quad \bar{\Delta}_E = \frac{S_+ - S_-}{|E|} \quad (10)$$

one has $D_E = |\Delta_E| + C_E$ with $C_E = 2\min(S_+, S_-)/|E| \geq 0$; C_E equals $2/|E|$ times the total magnitude of the changes whose sign is opposite to that of Δ_E ; and, if $D_E > 0$, $\kappa_E \in [0, 1]$, with $\kappa_E = 0$ if and only if no two changes have opposite signs and $\kappa_E = 1$ if and only if $\Delta_E = 0$.

Proof. Splitting the sums over E by the sign of Δ_e gives equation (10). For real $a, b \geq 0$, $a+b-|a-b| = 2\min(a, b)$, so $C_E = D_E - |\Delta_E| = 2\min(S_+, S_-)/|E| \geq 0$. If $\Delta_E > 0$, then $S_+ > S_-$ and $\min(S_+, S_-) = S_-$, the total magnitude of the changes of sign opposite to Δ_E ; the case $\Delta_E < 0$ is symmetric, and if $\Delta_E = 0$, then $S_+ = S_-$ and $C_E = D_E$. Dividing by $D_E = (S_+ + S_-)/|E| > 0$ gives $\kappa_E = 2\min(S_+, S_-)/(S_+ + S_-)$, which lies in $[0, 1]$ because $\min(a, b) \leq (a+b)/2$, vanishes exactly when $S_+ = 0$ or $S_- = 0$, and equals one exactly when $S_+ = S_-$, that is, when $\Delta_E = 0$. ■ The bounds stated in the main text follow from $|\Delta_e| \leq |\tau_b^{(e)}| + |B_e| \leq 1 + |B_e|$. A unit test checks the identity, the minority-mass form and the bounds on 20,000 random vectors, including exactly balanced, one-signed and tie-heavy cases, and the simulation checks them in every monitored window (largest absolute discrepancy 6×10^{-17}).

All pairs versus the edge set

Because B_p is a calibration mean for every $p \in P$, the change of the all-pairs Systemic Tau from its calibration mean is $\Delta_P = |P|^{-1} \sum_{p \in P} \Delta_p$. Splitting the sum over E and $P \setminus E$ gives the first part of equation (5), and hence $\Delta_P - \Delta_E = (1 - \rho)(\Delta_{P \setminus E} - \Delta_E)$. With $R = D_E - |\Delta_P| - C_E = |\Delta_E| - |\Delta_P|$ by Proposition 1, the reverse triangle inequality gives $|R| \leq |\Delta_E - \Delta_P|$, which is the bound. It held in every simulated window.

Invariances and power means

τ_b depends on the streams only through the signs $\xi_i(k, l)$, which strictly increasing transformations of z_i leave unchanged and strictly decreasing ones negate. A decreasing transformation of module i therefore negates $\tau_b^{(ij)}$ in every window, hence B_{ij} and Δ_{ij} , for every j ; $|\Delta_{ij}|$ is unchanged, so D_E is invariant, whereas Δ_E , and with it C_E and κ_E , changes unless no edge of E is incident to i or the changes of those edges sum to zero. The mean absolute change is the case $q=1$ of the power means

$$D_{E,q} = \left(\frac{1}{|E|} \sum_{e \in E} |\Delta_e|^q \right)^{1/q}, \quad D_E = D_{E,1} \leq D_{E,2} \leq D_{E,\infty} = \max_{e \in E} |\Delta_e|, \quad D_{E,2}^2 = \bar{\Delta}_E^2 + s_E^2 \quad (11)$$

which do not decrease with q . The quadratic member decomposes into the squared signed change and the across-edge variance s_E^2 of the changes, a second-order analogue of Proposition 1; the limit $q \rightarrow \infty$ is the largest single change, whose argument is the source e^* of equation (8). Larger q weights concentrated change more heavily, whereas $q=1$ keeps the decomposition additive on the scale of the changes themselves.

No-change behaviour

Without change, $E[\Delta_e] \approx 0$, and when the monitored window does not overlap the calibration windows and windows are nearly uncorrelated, $\text{Var}(\Delta_e) = \text{Var}(\tau_b^{(e)})(1 + 1/n_{\text{cal}})$. For serially independent streams, the variance of τ_b under independence, conditional on the ties in the window, is Kendall's⁵

$$\text{Var}_0[\tau_b] = \frac{V_s}{(n_w - a_1/2)(n_w - b_1/2)}, \quad n_w = \frac{W(W-1)}{2}, \quad (12)$$

with

$$V_s = \frac{W(W-1)(2W+5) - a_3 - b_3}{18} + \frac{a_2 b_2}{9W(W-1)(W-2)} + \frac{a_1 b_1}{2W(W-1)}, \quad (13)$$

where n_w is the number of bin pairs and the sums $a_1 = \sum_g t_g(t_g-1)$, $a_2 = \sum_g t_g(t_g-1)(t_g-2)$ and $a_3 = \sum_g t_g(t_g-1)(2t_g+5)$ run over the groups of t_g tied values of z_i in the window, and b_1, b_2, b_3 likewise for z_j . Without ties it reduces to $2(2W+5)/(9W(W-1))^4$, and for binary streams, for which τ_b is the φ coefficient, to $1/(W-1)$. Serial correlation inflates the variance; following Bartlett¹⁰, we use

$$\sigma_e^2 = \text{Var}_0 \left[\tau_b^{(e)} \right] \left(1 + \frac{1}{n_{\text{cal}}} \right) f_e, \quad f_e = 1 + 2 \sum_{k=1}^K r_i(k) r_j(k) \quad (14)$$

with lag- k autocorrelations $r_i(k)$ of the binned streams estimated from the calibration record and $K=10$. The unit tests compare equation (12) with 20,000 random permutations of tie-heavy count series and with the binary case. Because τ_b is asymptotically normal, $|\Delta_e|$ is approximately half-normal, with mean $(2/\pi)^{1/2} \sigma_e$ and variance $(1-2/\pi) \sigma_e^2$, which gives $E_0[D_E]$ in equation (6) and $\text{SD}_0[D_E] \approx ((1-2/\pi) \sum_e \sigma_e^2)^{1/2} / |E|$. For mutually independent modules the errors of distinct pairs are uncorrelated to first order: two pairs that share module i also involve modules $j \neq k$ whose sign processes are independent of each other and of i and have zero mean for stationary, time-reversible streams. Δ_E is then approximately normal with variance $\sum_e \sigma_e^2 / |E|^2$, which gives $E_0[|\Delta_E|]$, and the ratio of the two means gives κ_0 . By the Cauchy–Schwarz inequality $(\sum_e \sigma_e^2)^{1/2} / \sum_e \sigma_e \geq |E|^{-1/2}$, with equality when all σ_e are equal, so $\kappa_0 \leq 1 - |E|^{-1/2}$ to this order. When modules are coupled, errors of pairs that share a module are correlated, the variance of Δ_E is larger and κ_E without change is lower (0.74 in the coupled simulation against 0.75 for independent streams).

Sensitivity to coherent change

If $\Delta_e \sim N(\delta, \sigma^2)$ on every edge, $E|\Delta_e| = \sigma h(\delta/\sigma)$ with h as in equation (7); the expansion follows from $\exp(-u^2/2) = 1 - u^2/2 + O(u^4)$ and $2\Phi(u) - 1 = 2u\varphi(0) + O(u^3)$, with $\varphi(0) = (2\pi)^{-1/2}$. The expected excess of D_E over its no-change value is therefore $(2/\pi)^{1/2} \delta^2 / (2\sigma)$, against a null standard deviation $\sigma(1-2/\pi)^{1/2} |E|^{-1/2}$, whereas the signed change grows by δ against a null standard deviation $\sigma |E|^{-1/2}$. With $u = \delta/\sigma$, the standardized separations are $|E|^{1/2} u^2 / (2(\pi/2-1)^{1/2})$ for D_E and $|E|^{1/2} u$ for the signed control, so the signed control separates weak coherent change better; the leading-order expressions cross at $u = 2(\pi/2-1)^{1/2} \approx 1.51$. For balanced change the signed control has no separation at all.

Conformal alarm rule

Let windows $1, \dots, n_{\text{cal}}$ be the calibration windows and window $n_{\text{cal}}+1$ the monitored one. Each of the $n_{\text{cal}}+1$ windows receives the score

$$s_k = \frac{1}{|E|} \sum_{e \in E} \left| \tau_b^{(e)}(k) - \frac{1}{n_{\text{cal}}} \sum_{v \neq k} \tau_b^{(e)}(v) \right|, \quad k = 1, \dots, n_{\text{cal}} + 1 \quad (15)$$

where v runs over the other n_{cal} windows, so that $s_{n_{\text{cal}}+1} = D_E(t)$ exactly. Each score is a symmetric function of the other windows; if the $n_{\text{cal}}+1$ window matrices are exchangeable, so are the scores, the rank of the monitored score among them is uniform, and the p-value of equation (8) satisfies $\Pr(p_D \leq \alpha) \leq \alpha^{12}$. With $n_{\text{cal}}=39$ and $\alpha=0.05$, the alarm fires when $D_E(t)$ exceeds all but at most one

calibration score. Replacing the mean absolute deviation by the absolute mean deviation gives the signed controls on E and on P . Consecutive windows of a stationary process are only approximately exchangeable; in the simulations, latent states persisted for about 33 ticks against windows of 600 ticks, and the empirical false-alarm rates were within binomial error of α .

Simulation model and protocol

Module i emits an event at tick t with probability

$$\lambda_i(t) = \min \left(1, \max \left(0, p_i \left(1 + a \sum_{e \ni i} w_{ie} c_e(t) + \gamma g(t) \right) \right) \right) \quad (16)$$

where each edge e carries a latent state $c_e(t) \in \{-1, +1\}$ that keeps its value from one tick to the next with probability 0.97, $g(t)$ is a common latent state of the same kind, $w_{ie}=1$ for the first endpoint of e and $w_{je}=s_e \in \{-1, +1\}$ for the second, and $a=0.5$. The coupling graph is a ring of 10 modules with 6 random chords (16 edges, 8 with $s_e=+1$ and 8 with $s_e=-1$); the base probabilities p_i were drawn once from $U(0.08, 0.20)$ (range 0.104–0.186). Events are counted in bins of 10 ticks, and windows contain 60 bins. Each replicate draws 39 consecutive calibration windows, and each monitored condition draws 5 consecutive windows from a separate stretch of the process, with common random numbers across conditions. Balanced rewiring multiplies w_{je} by $1-2\theta$ on a set of four concordant and four anti-concordant edges ($\theta=1$ in Table 1), chosen in a 2,000-window pilot as the combination whose expected baselines sum closest to zero. The coherent shift sets $\gamma>0$; Table 1 uses $\gamma=0.5$, the value on the grid 0.1, 0.2, ..., 0.7 whose mean absolute expected edge change, the mean over edges of the absolute mean of Δ_e over monitored windows, is closest to that of rewiring at $\theta=1$ (0.104 against 0.119). The independent null drives each module by private copies of its latent states with the same weights, which preserves the marginal law and autocorrelation of each stream and removes all cross-dependence. Circular-shift surrogates rotate each module's concatenated calibration record, except that of a reference module, by an independent lag of at least one window. Detection curves use $\gamma \in \{0, 0.1, \dots, 0.7\}$ and $\theta \in \{0, 0.125, \dots, 1\}$. All random streams derive from one NumPy seed sequence (seed 20260930), and the full configuration is written with the results.

Data and code availability

No experimental data were used. The Python package that implements the operator, the decomposition, the null guide, the conformal rule and the simulation, together with its unit tests and a script that regenerates every number in the text, Table 1 and Fig. 1 with a single command, is archived on Zenodo alongside this note, with the resulting CSV and JSON files. The τ_b estimator follows the reference implementation of Systemic Tau¹⁷.

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Author contributions

J.P.-V. conceived the study, derived the results, wrote the code and wrote the manuscript.

Competing interests

The author declares no competing interests.

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